

Jamie Swaine

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EDUCATION

Imperial College London

MSc Geo-Energy with Machine Learning and Data Science – Current grade: Distinction 09/2025 - 09/2026

- Numerical Programming in Python (A), Computational Mathematics (A), Deep Learning (A), Fluids and Fluid Flow (A*), Energy Geoscience and Engineering (A*), Applying Computational and Data Science (A)

University College London

BSc Chemistry – First Class Honours 09/2022 - 06/2025

- Concepts in Computational Chemistry (~ 79%), Mathematics for Physics and Astronomy (~ 82%), Scientific Programming (~ 81%), Chemical Literature (~ 73%)

PROFESSIONAL EXPERIENCE

AWE Nuclear Security Technologies – Blacknest

Research project – Seismic Event Classification 05/2026 - 09/2026

- Developed an explainable ensemble (**PyTorch**, **Scikit-Learn**) – including a novel frequency-waveform-physics transformer – combined via a calibrated **MLP meta-learner**, achieving **0.98+ F1** on nuclear event discrimination
- Designed a **Viola-Jones-inspired anomaly-screening cascade** using generative models (**VAE**, **DDPM**) to triage high-volume classes ahead of the full ensemble, cutting downstream compute for >60% of traces
- Unified and preprocessed multiple heterogeneous global seismic time-series sub-archives (~**100GB**) into a single leakage-audited dataset spanning six event classes across >**400,000** traces
- Built a cross-architecture explainability suite (**integrated gradients**, **Grad-CAM**, **saliency**, **attention rollout**), confirming attributions consistently localise to physically meaningful source-physics regions at the population level
- Distributed model training and **Optuna** hyperparameter tuning across multiple GPUs on **HPC** clusters via **PBS** job arrays, leveraging **Claude Code** for autonomous refactoring and porting; tracked experiments with **MLflow**

AWE Nuclear Security Technologies

Research Intern – Generative Crystal Structure Discovery 06/2025 - 09/2025

- Trained and deployed a **conditional LLM** for crystal structure generation using **PyTorch** and **Hugging Face**, conditioning generation on structure properties
- Designed **novel conditioning vectors** combining statistical and thermodynamic descriptors, and utilised **Sun Grid Engine** to distribute large-scale model training and **hyperparameter optimisation** across GPUs
- Identified **three previously undocumented theoretical crystal structures**, confirmed stable at the **hybrid DFT** level, and presented findings to a multidisciplinary audience
- Curated and combined public datasets and developed **Bash** pipelines to address crystal structure class imbalance and orchestrate data processing workflows

Training Associates

Administrator 11/2020 - 09/2022

- Analysed **6+** years of financial and audit data, identifying departments at risk of funding reclamation
- Coordinated timetabling across **25** branches and **10+** staff, managing a **100%+** increase in healthcare training demand

PUBLICATIONS AND CONFERENCES

Swaine J., Bone C., Kayastha P., Butler K., Galloway E., Darby M. (2026)

- *Conditional Generative Models Enable Targeted Exploration of MAX Phase Design Space*
- Journal of Materials Chemistry C. **95th percentile novelty**, and **80th percentile impact**.

Swaine J., Green D., Moseley B. (2026)

- *An Explainable Ensemble for Multi-Class Seismic Event Discrimination*
- Manuscript in preparation

Ayivi W., Zhang X., Aligayev A., Kwao L., Ativi W., Sam F., Swaine J., Owusu S. (2025)

- *Brain-EffNet: A Lightweight and Explainable EfficientNet Framework for Multi-Class Brain Tumor Classification from MRI Images*
- Submitted to **Springer Nature – The Journal of Supercomputing**, pending review

Aligayev A., Jabbarli U., Swaine J., Jianlin L., Samadova U., Jastrzębska A., Domínguez-Gutiérrez F. J. (2025)

- *Vacancy-Confined Single-Atom Catalysts on MXenes for Enhanced CO₂ Reduction*
- Poster presented at **FEMS EUROMAT 2025**